

No.	Solution	Remarks
5(a)(i)	$V = JC^{-1}$ $= \text{kg ms}^{-2} \text{ m (As)}^{-1}$ $= \text{kgm}^2\text{s}^{-3}\text{A}^{-1}$	[1] correct substitution [1] correct final answer
5(a)(ii)	The potential difference (p.d.) between two points in a circuit is the <u>energy converted from electrical to other forms per unit charge</u> passing from one point to the other. On the other hand, electromotive force (e.m.f.) of a source is the <u>energy converted from other forms to electrical per unit charge</u> driven through the source.	[1] correct definition for p.d. [1] correct definition for e.m.f.
5(b)(i)	$I = \frac{E}{r + R_T}$ $P = I^2 r$ $= \frac{E^2 r}{(r + R_T)^2}$	[1] correct expression for I [1] for correct answer
5(b)(ii)	The power dissipated in the internal resistor <u>increases</u>. When <u>temperature increases</u>, the <u>resistance of the thermistor decreases</u>, causing the <u>current to increase</u>, resulting in power dissipated in the internal resistor to increase.	[1] state correctly [1] explanation
5(c)	Since wire X and wire Y are connected in series, the same current passes through both wires $I_X = I_Y$ $n_X A_X q v_x = n_Y A_Y q v_Y$ The drift velocity of charged carriers (electrons) in wire X is three times that of in wire Y, and the radius of wire X is half that of wire Y, therefore $n_X A_X q(3v_Y) = n_Y A_Y q v_Y$ $n_X \left(\frac{1}{4} A_Y\right) q(3v_Y) = n_Y A_Y q v_Y$ $\frac{n_Y}{n_X} = \frac{\left(\frac{1}{4} A_Y\right) q(3v_Y)}{A_Y q v_Y}$ $\frac{n_Y}{n_X} = \frac{\left(\frac{1}{4}\right)(3)}{1}$ $\frac{n_Y}{n_X} = 0.75$	[1] correct association of drift velocity and current for both wires [1] for correct substitution for drift velocity and cross-sectional area [1] for correct answer

6

No.	Solution	Remarks